

Kim Hanbyul 김한별

From biomedical engineering, semiconductors and medical AI to neuromorphic semiconductors — aiming to realize the principles of the brain as artificial-brain hardware.

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CURRENT POSITION

May 2026 – Jan 2027 (contract)	KIST · Research Intern — Neuromorphic Computing & Spiking Neural Networks SNN modeling and dynamical analysis of cerebellar circuits; plasticity-control algorithms under neuromodulators
	National R&D project · Participating researcher Development of original cerebellum-inspired semiconductor technology integrated with non-volatile memory for real-time predictive control of physical AI systems Next-Generation Intelligent Semiconductor Technology Development (Devices) — National R&D Program · RS-2026-25540328 PI: Dr. Jaewook Kim (KIST Center for Semiconductor Technology) · Participation May 1 – Dec 31, 2026 (8 months) · whole project Apr 2026 – Dec 2028

EDUCATION & TRAINING

Mar 2020 – Feb 2026 (graduated)	● Konkuk University GLOCAL Campus — B.S. College of Science and Technology — ICT Convergence Engineering / Biomedical Engineering ● Minor: Department of System Semiconductor Engineering Cumulative GPA 4.23 / 4.50 (percentile 97.3) · major 4.31 (98.1) · minor 4.38 (98.8) · 149 credits Academic honors 5 semesters — 2020-1 · 2020-2 · 2023-2 · 2024-2 · 2025-1
2025	● Medical AI — K-Digital Training (1,026 h) Future Convergence Education Center, Daejeon · Konyang University Hospital Biomedical AI program (K-Digital Training, 1,026 h)
Key coursework	Biomedical engineering Biosignal and SystemBiomedical Signal ProcessingDigital System for BMEInstrumentationFundamentals of Electronic Medical Devices and ExperimentsElectronic Circuit Laboratory for BME Semiconductor Memory Semiconductor DevicesSemiconductor EquipmentSemiconductor Test and Packaging ProcessBasic Semiconductor TestSemiconductor ChemistryQuantum Mechanics Computing Computer ArchitectureMicroprocessors & SOCComputer Programming

SCHOLARSHIPS

2024-1	★ National Scholarship for Science & Engineering (2 years)
2023-2 · 2024-1	Merit scholarship (Uijo)
2020-2 · 2023-2	Contest scholarship (national innovation fund)

PROJECTS

2024 – 2025	AI assistant on ESP32-S3-BOX-3 (capstone) Built an IoT framework — Capstone Design 1 · 2
2026	ai-research-harness An AI workflow scaffold that keeps the researcher in charge — format roles, glossary, approval gates, error log brain-cdss-daejeon Brain clinical decision support system (Brain CDSS) — JavaScript / Django

SKILLS

Biomedical Engineering	Medicine	Electronics	C/C++	IoT	Biosignal processing
Semiconductor	RISC-V	Semiconductor process	Computer architecture (ALU · memory)	Logic operations	
Medical AI	PyTorch	Statistics	Swin UNETR (MRI)	Genomic data models	PACS Orthanc

ACTIVITIES

2024.01 – 02	Overseas field program — New Frontier, USA
2025.01 – 02	Overseas cultural program — Chungbuk, Singapore
2025.01 – 02	Overseas cultural program — GLOCAL, Taiwan

MILITARY SERVICE

2021 – 2022	Military service — armor crewman · discharged
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